

Advanced Docker Training

Course Name	Advanced Docker Training
Course Description	Docker is an open-source project that automates the deployment of application inside software containers. Docker provides an integrated technology suite that enables development and IT operations teams to build, ship, and run distributed applications anywhere. Docker lets you run your software components on any platform that supports Docker containers.
Target Population	This course is for developers and IT operations teams who already use Docker and wish to acquire a deeper understanding of underlying infrastructure in order to improve the way they operate containerized workloads.
Pre-requisites	Basic experience building and running Docker containers. Basic Docker CLI knowledge. Docker for Mac/Windows will have to be installed during the workshop if working with Windows or MacOS. Prior experience in developing web applications will be helpful but is not required. Basic knowledge of Linux file system and network stack.
Course Topics	Containers and Micro services ○ Twelve-Factor App Concept Overview ○ How Docker facilitates Micro service Architecture ● Managing Docker images ○ Building images - best practices ○ Docker registries overview ● Container Data Management ○ Data Volumes ○ Data volume Containers ○ Filesystem Drivers ● Connecting Containers ○ Manually Connecting Containers ○ Connecting Containers using Docker Compose ● Docker network function deployment and configuration ○ Docker Network introduction ○ Network configuration and mapping ○ DNS and Bridge configuration ○ Port Mapping ○ Network advanced features ● Docker performance monitor and troubleshooting ○ Logging ○ Container performance monitor principles and tools ○ Docker runtime and cluster monitoring ○ Performance monitor system deployment principles

	● Docker Security ○ Docker security flow ○ Image Signing (Docker Notary & Content Trust) ○ Known vulnerabilities ○ Security scanning ● Container Schedulers/Orchestrators Overview ○ Docker Swarm ○ Kubernetes ○ Mesos ○ Nomad ● Continuous Integration and Deployment process using Docker ○ Jenkins Integrations ○ Self-contained builds with Docker containers ○ Docker multi-stage builds ● Using the Docker API ○ Remote docker hosts ○ Retrieving stats
Course Duration	2 Days (16 hours)